EXPERT OPINION

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An update on recent patents on thrombin inhibitors (2010 – 2013)

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Introduction: Thromboembolic incidences have increased nearly 33% in the past decade and directly affect nearly 0.5% of the population. Heparin, warfarin and current direct thrombin inhibitors (DTIs), the primary anticoagulants of choice, suffer from several drawbacks. Thus, the search for an antithrombotic devoid of adverse effect continues in earnest.

Areas covered: Literature search covering PubMed, SciFinder™ Scholar, Web of Knowledge, Espacenet, PatentScope and Google Patent Search was used to uncover > 35 patents describing new chemical entities and advances in DTI technologies. Our search uncovered considerable emphasis on the development of larger more complex molecules such as peptide-based inhibitors, prodrug derivatives, bivalent tryptophan zippers, triple action inhibitors and allosteric inhibitors. Advances in formulation technologies for clinically relevant DTIs have also been made.

Expert opinion: Thrombin is a multifaceted, dynamic enzyme with both coagulant and anticoagulant functions. Newer DTIs are attempting to fine tune thrombin's activity by targeting allosteric sites or by site-specific targeting of clotting. The complexity of thrombin's functions is driving the design of complex anticoagulants. Advancements in formulations and production processes have attempted to make traditional DTIs more cost effective to produce. The literature reveals a trend to develop a thrombin 'modulator' rather than an 'inhibitor.'

Keywords: allosteric inhibition, anticoagulant, coagulation enzyme, drug design, enzyme inhibition, thrombosis

Expert Opin. Ther. Patents (2014) 24(1):47-67

1. Introduction

The circulatory system has to constantly be on alert to maintain a delicate balance between blood flow and stasis under normal as well as abnormal conditions. Blood coagulation is the nature's defense mechanism that rapidly tips the balance between flow and stasis depending on the need [1]. Yet, this defense mechanism can be disrupted by several damaging internal cues that induce aberrant clotting. This results in the formation of a pathologic clot that can become life-threatening if not treated quickly. A pathologic clot, called thrombus, impairs normal blood flow, which manifests in the form of myocardial infarction, stroke, pulmonary embolism and many other conditions [2]. Together these arterial and venous thrombotic diseases are the leading cause of death in the world today.

Two major classes of drugs have been used in the past century to treat these thrombotic disorders. These include heparin and warfarin. Heparin, or more appropriately called unfractionated heparin (UFH), has been used as an anticoagulant in the clinic for over 90 years [3,4]. Even today, UFH and its recent derivatives including low-molecular-weight heparins (LMWHs) and heparin-like pentasaccharide (fondaparinux) are the most widely used anticoagulants in the world with an annual market cap of at least $10 billion. UFH, LMWHs and fondaparinux bind

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Article highlights.

- Thrombin, a highly plastic enzyme, is the key target for anticoagulant development.
- More than 35 patents have been issued in recent years describing new chemical entities and advances in DTI technology.
- Thrombin's complex functions are driving the design of complex DTIs.
- The future of newer DTIs may be to modulate, rather than inhibit, thrombin through allosteric or exosite specific targeting.

This box summarizes key points contained in the article.

antithrombin, a human plasma glycoprotein, to accelerate the inhibition of one or more enzymes of the coagulation cascade including thrombin, factor Xa and factor IXa (**Figure 1**). The inhibition is extremely fast and quantitative, which makes heparins very efficient anticoagulants. In addition, UFH is rather inexpensive and reversible. However, UFH and LMWHs are mixtures of considerable complexity that exhibit considerable off-target effects. A key consequence of these is significantly enhanced bleeding risk and the possibility of heparin-induced thrombocytopenia [5]. In addition, high response variability between patients and lack of oral bioavailability make heparins less than ideal anticoagulants.

The other traditional anticoagulant is warfarin (**1**), a vitamin K antagonist, which has been in use for > 50 years [3,4]. Mechanistically, vitamin K antagonists inhibit vitamin K epoxide reductase, a key enzyme involved in vitamin K cycle that is important for the generation of γ-carboxyglutamic acid residues in thrombin and other enzymes of the coagulation cascade. A major advantage of warfarin is that it is synthetically easy to produce and it is oral bioavailable. Yet, warfarin therapy suffers from its susceptibility to food–drug interactions causing serious bleeding complications and unpredictable pharmacokinetics, which requires constant monitoring [6].

The past decade or so has seen several new anticoagulants reach the clinic. These include argatroban (**4**), melagatran (**5**) (in the form of its prodrug ximelagatran (**6**)) and dabigatran (**7**) (in the form of its prodrug dabigatran etexilate (**8**)), which target the active site of thrombin and apixaban (**2**) and rivaroxaban (**3**), which target the active site of factor Xa [7,8]. Although ximelagatran was taken off the shelves soon after its introduction, these newer anticoagulants show good promise and have improved the safety profile of anticoagulant therapy. Yet, a common problem with these drugs is still bleeding complications.

1.1 Thrombin and the coagulation cascade

The coagulation cascade is nature's exquisite mechanism of converting soluble factors into an insoluble, but highly controlled, aggregate called a 'clot.' The inactive forms, or zymogens, of several enzymes that freely float in blood are activated in a sequential manner upon an initiating signal, such as injury (**Figure 2**) [9], resulting in formation of soluble fibrin monomers that polymerize to insoluble fibrin.

Thrombin, also known as factor IIa (fIIa), catalyzes the final step of converting fibrinogen to soluble fibrin as well as stabilizes the insoluble fibrin through generation of factor XIIIa from its zymogen [10]. In addition to these two key reactions, thrombin plays multiple roles to essentially single-handedly drive the cascade [11]. It activates several factors including factor V and factor VIII, which are essential for the formation of prothrombinase and tenase complexes, respectively [12]. Likewise, it has been suggested to activate factor XI [13]. This means that thrombin's feedback activation is crucial for rapid clot formation.

A second aspect of thrombin's prothrombotic role is its activation of platelets. Thrombin interacts with a number of platelet surface glycoproteins including glycoprotein Ibα (GPIbα) that helps platelet activation by cleaving protease-activated receptors (PAR)-1 and -4 on the platelet surface to induce aggregation [14]. Finally, while thrombin exhibits a primarily procoagulant role, it also exhibits an anticoagulant role. Under appropriate conditions, thrombin binds to thrombomodulin to form a complex, which activates protein C to give activated protein C that cleaves factors Va and VIIIa resulting in feedback inhibition (**Figure 2**) [15]. Overall, highly specific interactions with a multitude of protein counterparts drive thrombin's coagulant, platelet-activating and anticoagulant roles. These interactions can be modulated by targeting the specific domains on thrombin. A key modulator of thrombin function is heparin, which exploits exosite 2 to engineer specific recognition and fast inhibition [16].

1.2 The structure of thrombin

The three-dimensional structure of thrombin sheds light on how different substrates are specifically recognized (**Figure 3**). Thrombin, being a trypsin-like serine protease, contains an active site with Ser195, Asp102 and His57 (chymotrypsin numbering), forming the catalytic triad [17]. The active site of thrombin contains several subsites with characteristic features for substrate selectivity. The S1 specificity pocket, for example, is known to bind to electropositive arginine residue on fibrinogen and other substrates, and contains an aspartate residue (Asp189) at the base of the cleft. The S2 (proximal) pocket is more hydrophobic as compared to other serine proteases, while the S3 (distal) pocket is slightly acidic with a glutamate at position 192. The S4 pocket is a hydrophobic region with Try215 at its base [18]. A site located ~ 15 Å from the active-site catalytic triad and made up of a loop connecting two beta-strands is the sodium-binding site, which is known to be coupled thrombin's catalytic activity and selectivity of substrate recognition [19]. Occupancy of this site by sodium ion shifts thrombin from a slow to a fast form, which hydrolyzes substrates better [20,21]. In addition, there are two electropositive allosteric sites called anion-binding

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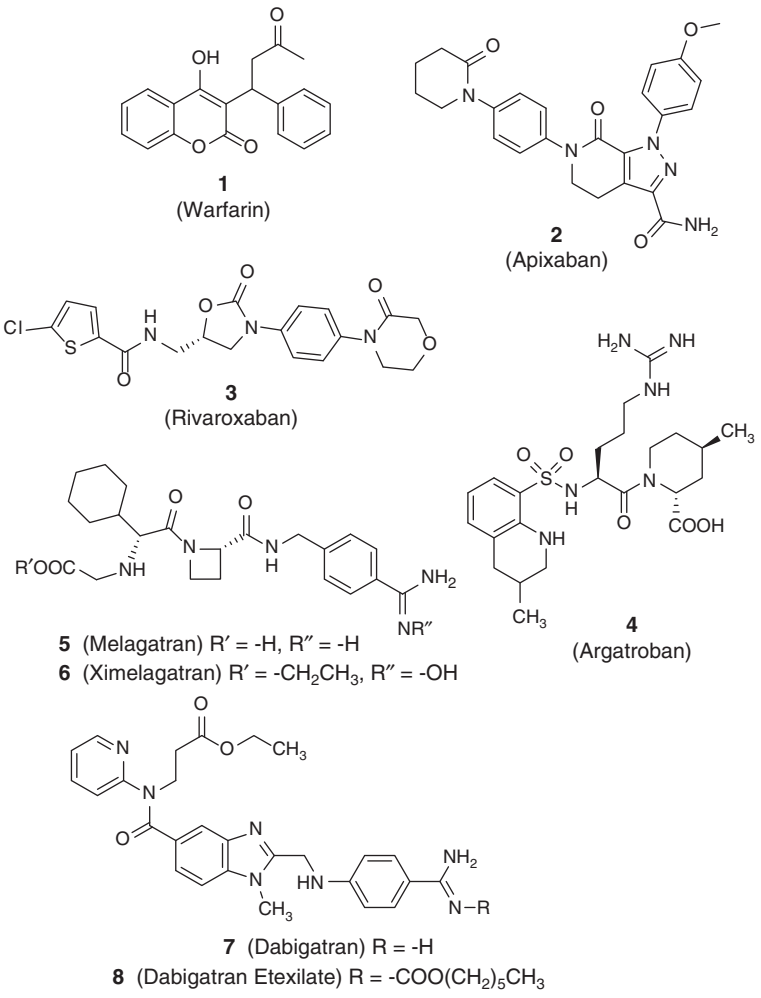


Figure 1. Small-molecule anticoagulant drugs (or their prodrugs) introduced in the clinic. Except for **6** (and hence **5**), all other drugs (or their prodrugs) are in clinical use today.

exosites-1 and -2, which bind a number of proteins. Exosite 1 is slightly less electropositive than exosite 2 and binds fibrinogen, which is essential for its rapid cleavage along the procoagulant pathway. Likewise, interaction with PAR-1, hirudin and heparin cofactor II utilizes exosite 1. Exosite 2, on the other hand, binds to more anionic species such as heparin, chondroitin sulfate of thrombomodulin and the sulfated tyrosines of GPIbα. Thrombin is a highly plastic enzyme that exhibits considerable ligand-dependent allosterism [22]. Although the debate on the crosstalk between the two exosites has not been settled, it is clear that both exosites induce conformational changes in the active site resulting in modulation of thrombin's specificity and catalytic activity [23-29].

1.3 Thrombin inhibition

Thrombin is a well-established target for anticoagulant therapy. The fundamental rationale in inhibiting thrombin has been that it is the final enzyme of the coagulation cascade that represents the culmination of the procoagulant signal. Major efforts in the past decade have resulted in at least three clinical viable candidates, argatroban (**4**), ximelagatran (**6**) and dabigatran etexilate (**8**). Of these, ximelagatran was withdrawn due to significantly enhanced hepatotoxicity [30]. Yet, the search for a better anticoagulant continues. **Figure 4** showing the state of pursuit of thrombin inhibitors over the past seven years highlights the continuing effort being invested across academia as well as industry to improve thrombin inhibition based anticoagulant therapy.

2. New chemical inhibitors of thrombin

Small direct thrombin inhibitors (DTIs) have gained significant ground over the past decade as an alternate means of antithrombotic treatment. The first injectable small-molecule inhibitor of thrombin was approved by the Food and Drug Administration (FDA) in June 2000 called argatroban (**4**) (Pfizer) for treatment of thrombosis in patients with heparin-induced thrombocytopenia [31]. However, it took

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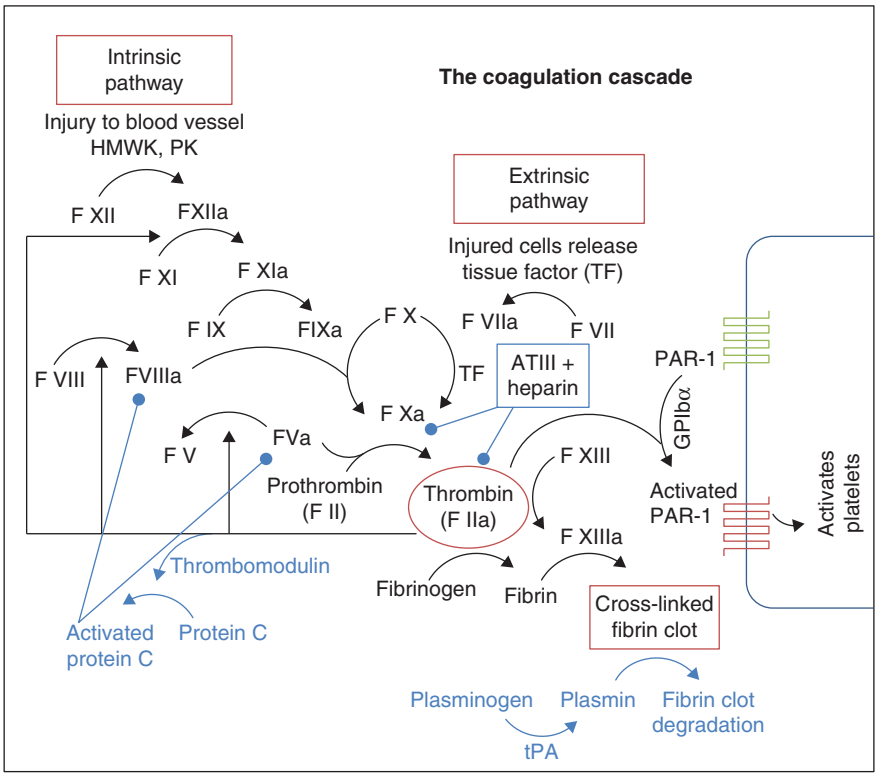


Figure 2. An overview of the coagulation cascade. In black are the procoagulant factors, while in blue are the anticoagulant factors. Thrombin (factor IIa) plays a key role in the cascade and acts as a link between the enzymes and platelets. It also provides negative feedback via its activation of protein C.

ATIII: Antithrombin; F: Factor; HMWK: High-molecular-weight kininogen; PAR: Protease-activated receptor; PK: Prekallikrein.

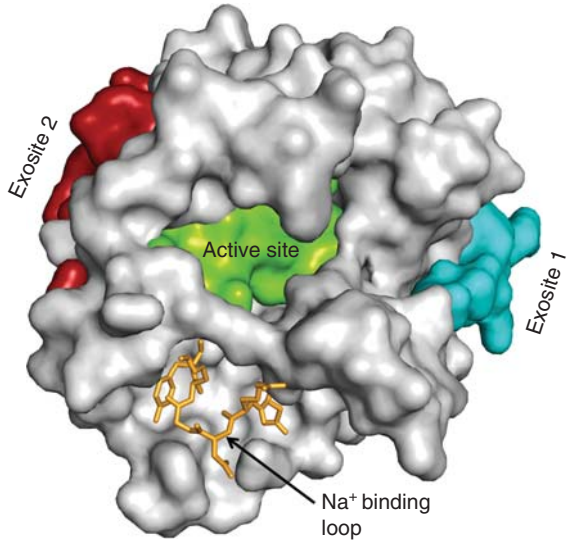


Figure 3. A crystal structure of thrombin (PDB ID: 1PPB) highlighting the active site in green, exosite 1 in teal, exosite 2 in red and the Na^+ binding loop in orange.

another 10 years for the development and approval of orally bioavailable inhibitor, dabigatran etexilate (**8**) (Pradaxa, Boehringer Ingelheim) [32]. The ideal thrombin inhibitor should have high enzyme inhibitory potency, high selectivity, *in vitro* and *in vivo* antithrombotic efficacy, a long half-life and oral bioavailability to allow once- or twice-daily dosing [33]. Dabigatran achieves most of these aims and the recent European Society of Cardiology's practical guide discusses its use for nonvalvular atrial fibrillation, which could result in stroke. It is the first direct thrombin inhibitor to be incorporated in any antithrombotic regimen [34]. Yet, it still suffers from some bleeding complications and has also been associated with altered renal function [35-38]. This implies that new thrombin inhibitors are of interest.

2.1 Peptide and peptidomimetic inhibitors

A major class of DTIs is peptide or peptide-derived small molecules. Starting with the design of peptide inhibitors on the basis of the natural substrate fibrinogen, which are generally basic in nature, today's inhibitors have evolved to be non-basic, which is a massive advance in terms of oral

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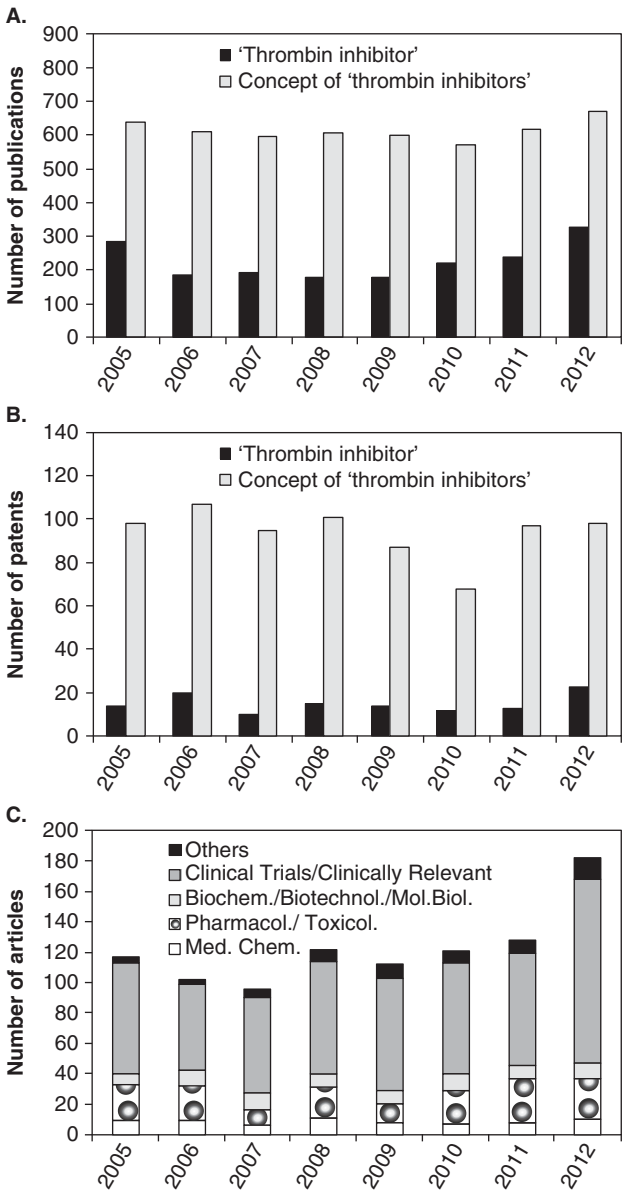


Figure 4. Longitudinal survey of SciFinder Scholar using 'thrombin inhibitor' as the search term. (**A**) The number of research publications (**A**) or patents (**B**) containing either the exact keyword 'thrombin inhibitor' or the concept of 'thrombin inhibitor.' SciFinder Scholar search engine sorts retrieved entries into 'high relevance' and 'some relevance,' which are described as either 'thrombin inhibitor' or concept of 'thrombin inhibitor.' An example of these two classifications is an entry that directly describes the development of a direct thrombin inhibitor ('most relevant') in contrast to an entry describing enzyme inhibitor design which may mention thrombin as one of the target enzymes. (**C**) Classification and number of article types into different categories using Web of Knowledge Science Citation Index. The search was performed for two keywords 'thrombin inhibitors' and 'factor IIa inhibitors,' combined and filtered according to different subcategories for specific years. The numerous subcategories listed by Science Citation Index were combined into a Medicinal Chemistry-related (Med. Chem.), Pharmacology-related (Pharmacol./Toxicol.), Biochemistry-related (Biochem./Biotechnol./Mol.Biol.), Clinical Trials-related (Clinical Trials/Clinically Relevant) and Others to ease interpretation. *Note*: > 70% of articles under Med. Chem. were about Medicinal Chemistry but also contained articles on analytical chemistry, organic chemistry, physical chemistry, and multidisciplinary chemistry. Likewise, articles under Pharmacol./Toxicol. also included health care sciences services, health policy services, and experimental medicine research, while Biochem./Mol. Biol./Biotechnol. also included biochemical research methods, biology, biophysics, applied microbiology, cell biology, crystallography, and physiology. Clinical Trials/Clinically Relevant retrieval items included an extensive group of fields ranging from cardiac and cardiovascular systems to transplantation and virology. Finally, others contained retrieval items in many numerous fields including agricultural engineering, engineering biomedical, evolutionary biology, materials science biomaterials, nanoscience nanotechnology, and veterinary sciences. It is also important to note that many articles are listed in multiple categories because of their diverse information content. For example, an article about drug design and its biochemical mechanism is classified under both Med. Chem. as well as Biochem.

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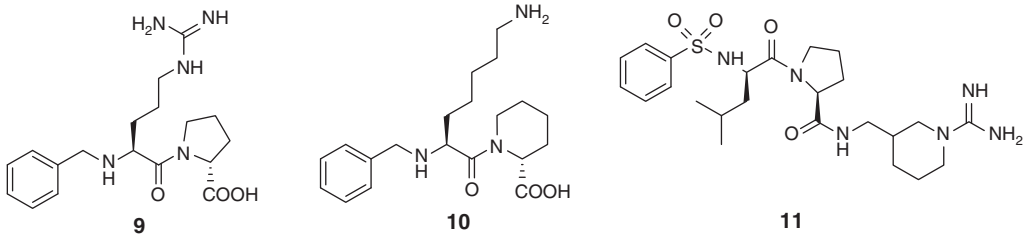


Figure 5. New dipeptide inhibitors of thrombin.

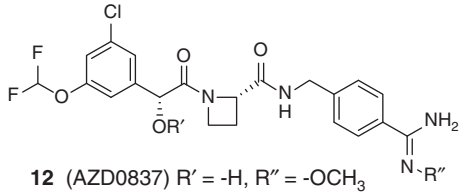


Figure 6. New tripeptide inhibitors of thrombin.

bioavailability. Whether basic or not, the majority of peptide or peptide-derived inhibitors compete with the substrate for the active site of thrombin.

2.1.1 *Dipeptide inhibitors*

To ensure affinity and potency, binding to at least two pockets on thrombin is required for direct inhibition. Shanghai Institute of Pharmaceutical Industry patented a series of simple, synthetic peptides containing a dipeptide backbone [39,40] in which Lys/Arg and Pro/Pip are the P1 and P2 residues, respectively, for example, as in **9** and **10** (**Figure 5**). Adding a hydrophobic amino acid as the P3 residue decreased the IC_{50} of thrombin inhibition to 100 nM or less, of which **11** (**Figure 5**) was the most potent inhibitor with an IC_{50} of 8 nM.

2.1.2 *Tripeptide inhibitors*

Based on the original Phe–Pro–Arg tripeptide [41], several potent DTIs have been developed. AstraZeneca developed melagatran (**5**), which in the prodrug form, ximelagatran (**6**), was the first oral anticoagulant since the introduction of warfarin. Although severe hepatotoxicity led to the withdrawal of ximelagatran, its success in prevention of stroke following nonvalvular atrial fibrillation and in subduing venous thrombosis following major orthopedic surgery still encourages tripeptide-based DTI development [42]. By modifying melagatran, AstraZeneca has discovered AZD0837 (**12**) (**Figure 6**), which has completed Phase II trials [43] (*Clinical Trials Identifier:* NCT00623779, NCT00684307) [44,45]. AZD0837 is orally bioavailable and is rapidly metabolized to its active form AR-H067637 (**13**) (**Figure 6**), which is a reversible DTI with an IC_{50} of 4 nM [46]. Clinical trials have shown that AZD0837 is effective at doses of 300 mg/day to treat patients with nonvalvular atrial fibrillation and was comparable to vitamin K antagonist warfarin [47-49]. However, the oral bioavailability of the drug varied greatly between 22 and 52%. An extended release formulation of AZD0837 is being evaluated in Phase I trials (*Clinical Trials Identifier:* NCT00878618) [50]. No study results have been posted to date. Tianjin Jiahong Technology LLC has patented prodrugs of AR-H067637 to improve the efficacy and oral absorption [51]. An ester prodrug of AR-H067637, that is, **14** (**Figure 6**), is suggested to be more lipophilic, and hence possibly more orally bioavailable, than AZD0837. More tests are necessary to support this hypothesis.

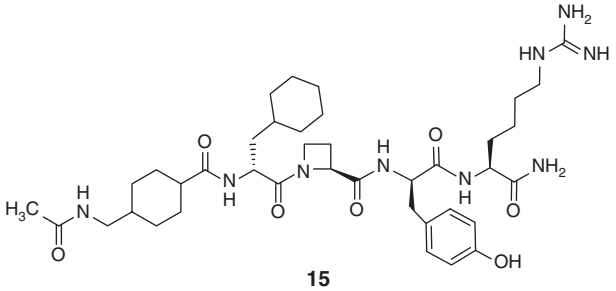
2.1.3 *Large peptidomimetic inhibitors and s-variegin peptide derivatives*

In conjunction with small substrate-like peptide DTIs, pharmaceutical industry has attempted to pursue the design of novel molecules. Matrix Pharma has utilized a powerful computer-assisted multiparameter optimization process (CADDIS) to design a series of new, reasonably large, peptide entities [52,53], while taking into account the affinity, solubility, stability, toxicity and selectivity of molecules. The new peptidomimetic DTIs exhibits inhibition constants (K_i) in the low nanomolar range. The X-ray structure reveals that the *D*-tyrosine of **15** (K_i = 3 nM) (**Figure 7**) forms a hydrogen bond with Asp189 of thrombin, while the rest of the molecule orients away from Ser195, the catalytic residue [54].

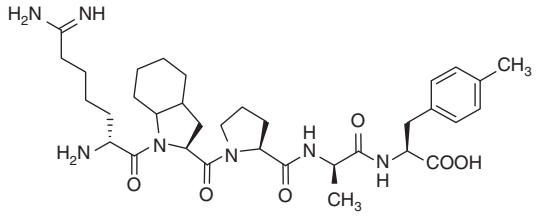
Because thrombin interacts with several proteins, a key paradigm is to target the interface between the proteins so as to generate a highly specific inhibitor. Bradykinin (Arg–Pro–Pro–Gly–Phe–Ser–Pro–Phe–Arg) is an active peptide released from plasma kininogens, which inhibits thrombin-induced platelet activation [55,56]. Bradykinin is a weak thrombin inhibitor (K_i = 170 – 326 μM) and prevents platelet activation by blocking the thrombin-PAR-1 interaction [57]. A series of pentapeptide bradykinin analogs have been developed by University of Michigan that show improved activity against platelet activation and PAR-mediated calcium mobilization [58]. The optimized compound **16** (**Figure 7**) inhibits α-thrombin-induced platelet aggregation 100% at 8.4 μM, while it inhibits thrombin with a K_i of 4.4 μM. The cocrystal structure with thrombin shows that **16** inserts into the active

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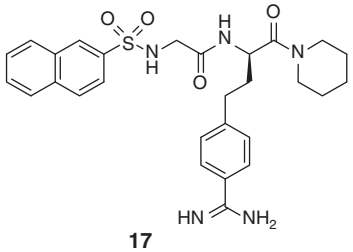


15



16

Figure 7. Large peptidomimetic inhibitors of thrombin.



17
(NAPAP)

Figure 8. Active site inhibitor NAPAP.

site in a retro-orientation by interacting with Asp189 and Gly216 [59].

Many peptide-based DTIs have been obtained from nature. The classic case is that of hirudin, which is obtained from the leech *Hirudo medicinalis*. Hirudin, and its less-potent derivatives, have been useful for patients who are refractory to heparin for the past 14 years [60]. Hirudin (K_i = 20 fM) inhibits thrombin by binding to both exosite 1 as well as the active site. Recently, a 32-residue peptide, named as variegin, was isolated from the tropical bont tick (*Amblyomma variegatum*) and found to inhibit thrombin by a mechanism similar to that of hirudin [61]. The synthetic form of variegin is called s-variegin. In 2010, the Natural Environment Research Council of Great Britain obtained a patent for modified s-variegins with equal and better potency towards thrombin [62]. The researchers suggest that s-variegin inhibits thrombin by disrupting its charge relay system.

Subtle modifications, such as addition of a sulfated tyrosine to improve the affinity for exosite 1, produced a variant peptide called DV24*K10RY*^{sulf} (sequence: 'SDQGDVAEPRMH KTAPPF-DFEAIPEEY^{sulf}LDDE'), which inhibited thrombin with an IC_{50} of 1.4 nM and a K_i of 42 pM. This potency overshadows the FDA-approved hirulog-1/bivalirudin peptide inhibitors by 70-fold (K_i = 2.3 nM). DV24*K10RY*^{sulf} was also found to inhibit thrombus formation for over 150 s in an *in vivo* zebrafish larvae circulation model at a dose of 10 nl of 500 µM (control of PBS showed thrombus formation at 19 s) [63]. At the same dose the hirulog-1 was able to prevent thrombus formation for 45 s, thus demonstrating even better efficacy for prevention of thrombosis than presently marketed drug. It is possible that these s-variegin derivatives may yield better anticoagulants in the future to replace traditionally used hirudin analogs.

2.2 Dabigatran-based inhibitors

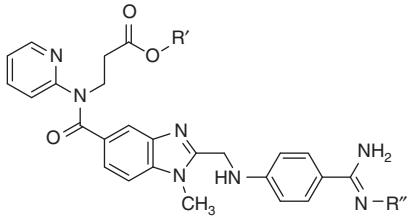
Dabigatran etexilate (**8**) (Boehringer Ingelheim code BIBR 953) is a clinically approved oral DTI for prevention of stroke in patients with atrial fibrillation. This diester prodrug is converted to the active form, dabigatran (**7**), by plasma esterases [64]. Dabigatran was designed from another DTI, NAPAP (**17**) (**Figure 8**). Introducing a benzimidazole at central carbon improves hydrophobic interaction within the proximal (S2) pocket, while the amidine forms a salt bridge with Asp189 and the pyridine ring fills the distal (S3) pocket [65].

Dabigatran etexilate has an oral bioavailability of ~ 7% [66,67]. Several groups have been pursuing structural modification to improve its bioavailability as well as effect faster conversion to the active form. Beijing Meibeita Pharmaceutical has patented a series of dabigatran esters [68,69]. Representative DTIs **18** and **19** (**Figure 9**) when given to mice at a dose of 10 mg/kg orally, show a 2.5-fold increase in the activated partial thromboplastin time (aPTT) and a 1.5-fold increase in mouse tail bleeding time. Prelude Pharm has developed a more potent dabigatran ester (**20**) (**Figure 9**) with twofold higher oral bioavailability [70]. It is converted into the active form faster and exhibits twofold longer half-life as compared to dabigatran etexilate. Tianjin Institute of Pharmaceutical Research has developed **21 – 23** (**Figure 9**) that are almost twice as potent as dabigatran etexilate [71-73]. Likewise, Suzhou Borui Pharmaceuticals has developed a series of dabigatran esters (**24**) (**Figure 9**); however, no activity data have been reported [74].

Rather than esterifying dabigatran, China Pharmaceutical University has incorporated other anticoagulant or cardiovascular-active entities in dabigatran [75]. Ligustrazine (tetramethylpyrazine) (**25**) (**Figure 10**) is an active component from traditional Chinese herbal medicine *Chuanxiong* rhizome. Ligustrazine has been proposed for the treatment of coronary atherosclerosis and ischemia [76-78]. Molecules containing both ligustrazine and dabigatran showed better antithrombotic profiles. For example, **26** (**Figure 10**) showed stronger inhibition of thrombin-induced platelet aggregation (IC_{50} = 218 nM) than dabigatran etexilate (IC_{50} = 326 nM) and **27** (**Figure 10**) demonstrated a promising effect in a venous thrombosis model (ED_{50} = 1.8 vs 4.4 mg/kg). Likewise, isosorbide-5-mononitrate (**28**) (**Figure 10**) is a well-established NO donor used in the treatment of angina

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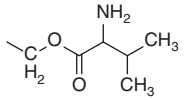
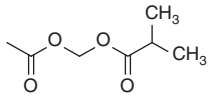
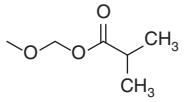
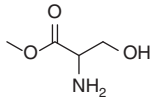
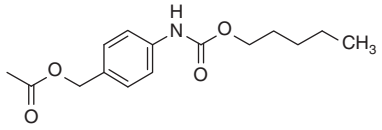
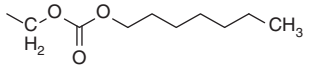
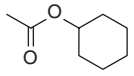
Compound no.	R′	R″
18	-CH_3	
19	-CH_2CH_3	
20	-CH_2CH_3	
21	-CH_2CH_3	
22	-CH_3	
23	-CH_3	
24	-CH_2CH_3	

Figure 9. Dabigatran ester derivatives.

pectoris [79]. A patent by China Pharmaceutical University combines isosorbide-5-mononitrate with dabigatran etexilate, that is, **29** (**Figure 10**), to improve vasodilation and inhibit thrombin-induced platelet aggregation (IC_{50} = 87 nM) [80].

Several groups have developed dabigatran analogs to improve DTI potency. China Pharmaceutical University has developed **30** that restricts the central linker through the formation an additional ring [81], which shows good inhibition of thrombin-induced platelet aggregation (IC_{50} = 159 nM). Fudan University modified the amidine group to guanidyl so as to strengthen the salt bridge to Asp189 as in 31 (**Figure 11**) [82]. Although a ninefold increase in clotting time (from 20 s to 178 s) was observed for **31**, no details of the assay have been described.

2.3 Nonpeptide inhibitors

The Institute of Oceanology of Chinese Academy of Sciences has discovered a potent thrombin inhibitor by screening biologically active compounds from marine wildlife [83,84]. This isobenzofuran-containing dibenzyl bromophenol compound (**32**) from the brown alga *Leathesia nana* exhibited significant thrombin inhibitory activity (IC_{50} = 6.87 nM) [85]. Despite the interesting activity, the binding mode of 32 appears to have not been studied to date. Also, studies on selectivity and toxicity are necessary to evaluate this new interesting class of DTIs. Similar bromophenol compounds have shown activity for/against other targets such as protein tyrosine phosphatase 1B [86,87], while possessing antitumor properties as well [88].

Bionika LLD built up a virtual library of molecules following the orcinol-based DTIs, which had shown good potency for thrombin with K_i in the nanomolar range [89-95]. To improve on the design, the virtual library was developed on the basis of an A–B–C-type substructure, which met the requirements for binding in thrombin's active site. The molecules mainly retained the benzenesulfonic acid (A) and orcinol (B) moieties, which were expected to bind in the S3 and S2 hydrophobic pockets. Group C of the molecules contained a positive charge so as to form a salt bridge with Asp189 at the bottom of the S1 pocket. Several molecules were selected through docking and scoring, of which **33** (IC_{50} (IC50 = 130 nM, Figure 12)= 130 nM) was reported as most potent in the patent. Lead **33** contained the phenylsulfonyl-like group of argatroban and the amidine group of dabigatran [96,97].

2.4 Bivalent tryptophan zipper containing direct thrombin inhibitors

The local temperature near a thrombus or atherosclerotic plaque might be 1 – 3°C higher than bulk blood temperature [98,99]. The exact reason for this temperature difference is not known, although it has been suggested to arise from either the higher levels of activated macrophages near the plaques or higher local blood viscosity and/or shear stress. Regardless of the factor(s), the temperature difference can theoretically be exploited for localized antithrombotic action.

An invention disclosed in 2012 presents a locally activated bivalent DTI, which is inactive in bulk blood but active near the site of thrombus [100]. The application discloses two nonoverlapping thrombin-binding peptides, which bind to the active site and the exosite 1, that are linked together through a tryptophan zipper (TrpZip or wZIP) sequence. TrpZip is a peptide sequence that exists in a β-hairpin configuration at lower temperatures, but denatures at higher temperatures to create a longer linker [101]. Thus, at normal body temperatures, the folded TrpZip would be noninhibitory because the two thrombin-binding domains would be only 4 – 6 Å apart, a distance not sufficient for optimal active site and exosite 1 interactions. Near a thrombus, the higher temperature would shift the TrpZip equilibrium more towards the unfolded state (span of 15 – 16 Å) resulting in an exponential increase in efficacy of inhibition (**Figure 13**). The example cited in the patent describes a peptide sequence in which the active site binding domain (IRFTD) is from hemadin, while exosite 1-binding

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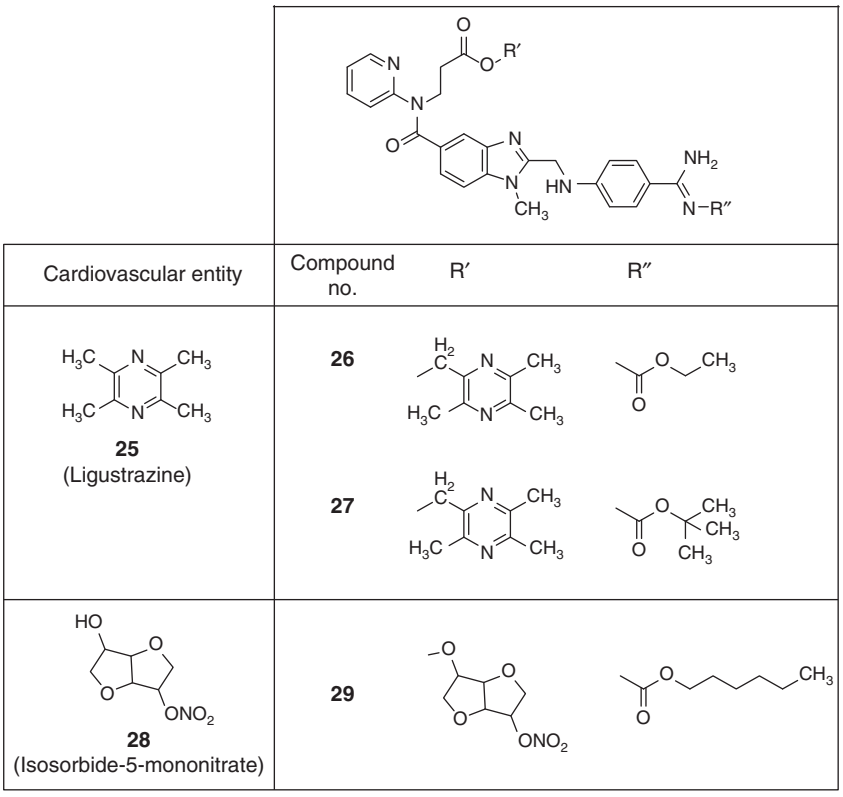


Figure 10. Multiaction dabigatran derivatives.

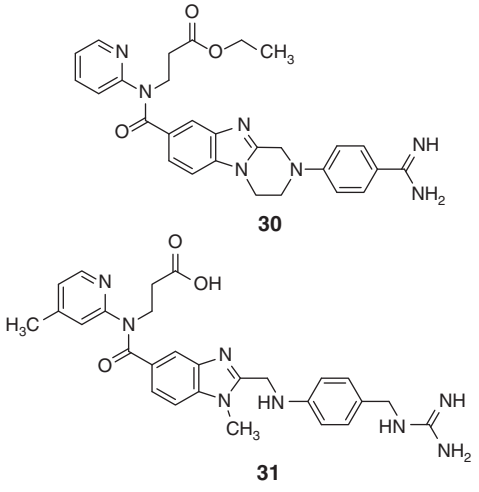


Figure 11. Dabigatran mimics.

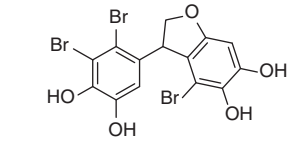
domain (GDFEEIPEEYLQ) is from modified hirudin. The two domains are linked together by a TrpZip sequence (GEWTWDDATKTWTWTE) to form MH2-wZIP4, which demonstrated better *in vitro* potency at higher temperature (IC_{50} = 245 nM at 25°C, IC_{50} = 148 nM at 42°C). In venous thrombosis model, MH2-wZIP4 (single bolus dose of 0.2 mg/kg) was particularly efficacious in reducing the weight of thrombus compared to 115 U/kg heparin, 1.5 mg/kg argatroban or 1.3 mg/kg bivalirudin. In arterial thrombosis model, a single bolus dose (0.2 mg/kg) of MH2-wZIP4 was not sufficient to observe any anti-thrombotic effect; however, supplementing the bolus dose with drug infusion (4.3 ng/kg/min) increased the efficacy significantly. It is likely that the higher pressure under arterial conditions necessitated continuous infusion of MH2-wZIP4. Despite this significant antithrombotic effect, the aPTT for fresh blood drawn from the test venous thrombosis rats was within the limits of the control group. Thus, the systemic effects of MH2-wZIP4 were little to none and that the drug was efficacious near the thrombi alone. Yet, higher doses of drug were not tested and questions on topical temperature increases causing any unwanted side-effects remain.

2.5 Multiaction inhibitors

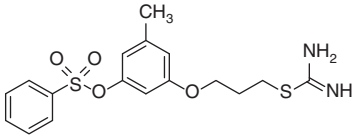
A dual-activity concept of combining heparin pentasaccharide, which activates antithrombin to accelerate factor Xa inhibition, with a DTI has been around since 1999 [102,103]. The idea is attractive because it entails simultaneous knock out of two key enzymes of the coagulation cascade. Yet, a dual thrombin and factor Xa inhibitor would completely abolish all hemostatic ability and may induce internal bleeding. Thus arose the triple activity concept, which enables reversal of bleeding using an antidote that recognizes the third

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32



33

Figure 12. Small nonpeptide inhibitors of thrombin.

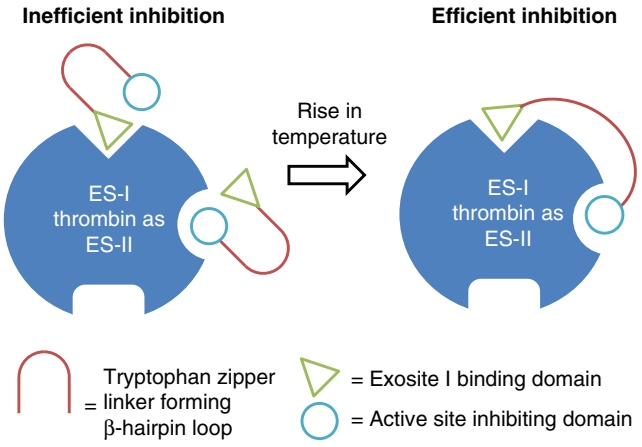


Figure 13. Exploiting the temperature differential between bulk blood and that near a clot for localized anticoagulant effect. TrpZip is a tryptophan zipper sequence containing an exosite 1-binding domain and an active site-binding domain at its two ends. At lower temperature, the folded TrpZip is incapable of binding to both sites resulting in inefficient inhibition, while the slightly higher temperature near a clot or plaque induces unfolding and subsequent binding to both the active site and exosite 1 on thrombin resulting in more potent inhibition.
ES-I: exosite 1; ES-II: exosite 2; AS: active site.

component present on the dual-targeting inhibitor. Endotis Pharma has developed such a triple action anticoagulant EP217609 (**34**) (**Figure 14**) by linking the heparin pentasaccharide with an α-NAPAP-like DTI and a biotin handle [104]. The α-NAPAP handle of EP217609 inhibits thrombin with very high affinity (K_i = 30 – 40 pM) and > 1000-fold selectivity over other serine proteases. The heparin pentasaccharide part of EP217609 binds rapidly to antithrombin with high affinity (K_D = 30 nM, k_{ass} ~ 10^6 $M^{-1}s^{-1}$) and activates it to selectively inhibit factor Xa over other coagulation proteases [105]. It has also been demonstrated that avidin can reduce the *in vivo* anticoagulant effect (20 – 100-fold) by binding to the biotin handle of the molecule [106] and can help with its clearance from circulation. Phase I clinical trials assessed the pharmacokinetic (PK), pharmacodynamic (PD), safety and tolerability of **34** [107]. This revealed no drug-related adverse effects or change in liver function > 10 days. The pharmacodynamics of EPI217609 showed that anticoagulant activity lasted for 72 h after single parenteral dose of 10 mg. Further, the PK/PD was very predictable with low intersubject variability. Although the original patent for EP217609 was for its use in ischemia-reperfusion injury during organ transplantation, it is currently being investigated in Phase II trials as a reversible anticoagulant for cardiopulmonary bypass surgery [105,108].

While the above example is essentially a macromolecule, a smaller molecule that is orally bioavailable and having dual inhibition would be of great advantage to clinicians. Verseon Corp. has patented a large number of substituted pyrazolyl and triazolyl derivatives with nanomolar IC_{50} [109] of these, two molecules (**35** and **36**) (**Figure 15**) were found to be dual inhibitors of both thrombin and factor Xa with IC_{50} of < 100 nM [110]. Interestingly, these molecules are small (MW < 600) and likely to be orally bioavailable. Further studies under *in vivo* conditions and also selectivity against other serine proteases would be important clinical relevance.

2.6 Allosteric inhibitors

2.6.1 Aptamers

Aptamers are small sequences of nucleic acids that are capable of binding a target protein or nucleic acid with high affinity [111]. Such aptamers can be useful in therapeutics, diagnostics or research. Using 'Systematic Evolution of Ligands by Exponential Enrichment' (SELEX), the first nucleic acid-based thrombin inhibitory aptamers were obtained by Bock *et al.* [112]. In this study, 32 thrombin-binding aptamers (TBAs) from a pool of DNA containing random nucleotide sequences contained a highly conserved 14—17 base region. Among these, the 15-mer called TBA_{15} (5′GGTTGGTGTGGTTGG3′) had a potency of 25 nM and was found to bind to exosite 2 [113,114]. X-ray and NMR studies of the TBA_{15}—thrombin complex has shown that the aptamer forms a G-quadruplex with a chair-like conformation consisting of two G-quartets, which stack one on top of another, and a TGT-loop, which connects the two [115-120]. Studies have also confirmed that a cation, such as K^+, is essential to improve the stability of the two G-quartets [121]. The novelty and potency of the TBA_{15} aptamer encouraged Archemix Corp. and Nuvelo, Inc., to conduct Phase I trials on their molecule ARC183, which was patented in 2007 [122]. However, due to the short *in vivo* half-life (~ 2 min) [123], high doses of aptamer were required to obtain any substantial effect, which resulted in suboptimal dosing profile and forced the companies to not perform further studies [111].

A recent European patent was granted to Apto-Farm for forming a relatively stable aptamer which could be more

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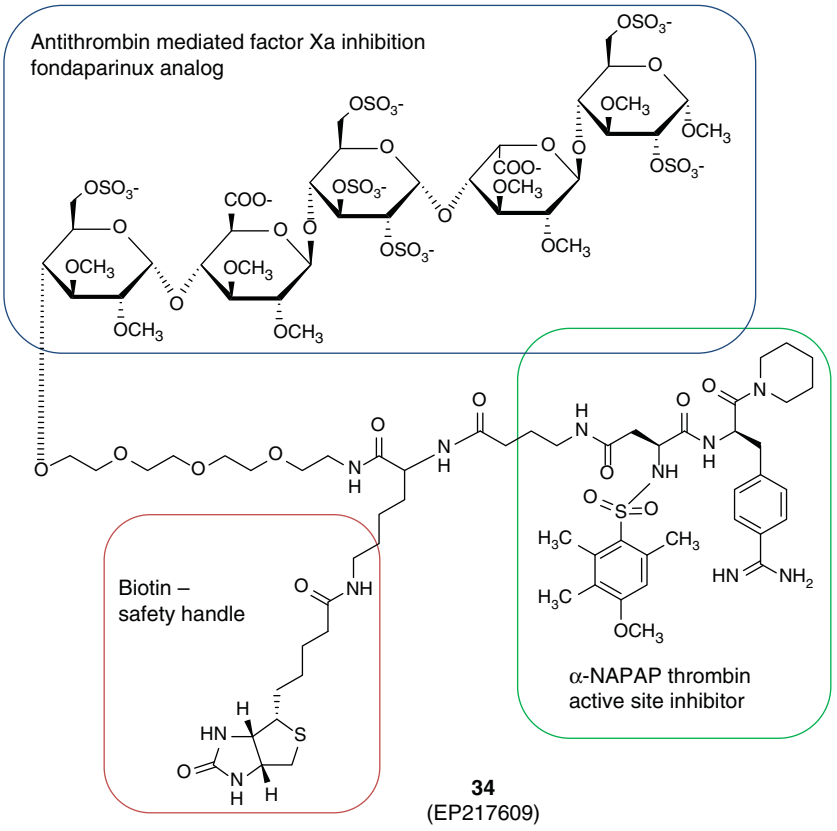


Figure 14. Shows the structure of EP217609 (34) containing α-NAPAP, heparin pentasaccharide and biotin, which bind to thrombin, antithrombin and avidin, respectively.

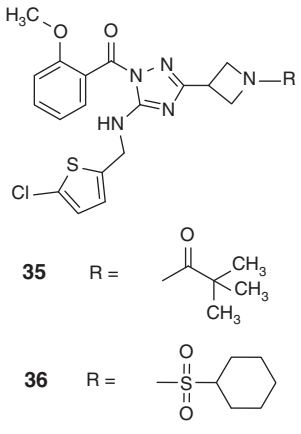


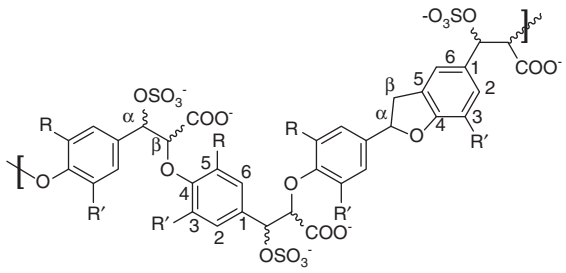
Figure 15. Small molecule dual inhibitors of fXa and thrombin.

therapeutically useful [124]. Apto-Farm has found that the stability of the TBA_{15} can be increased by increasing the stacking interactions in the upper quartet of the quadruplex and shielding the heterocyclic bases from water. Also, the stacking interactions along with steric hindrance restricted exchange of K^+ with the medium, thus enhancing stability. The patent claims that one of the aptamers DTI-33 (having sequence: 5′GGTTGGTGTGGNTGG-T_X-GGTTGGTGTGGNTGG3′, where N is any nucleotide and T_X is one or more bases of T) was active in rat blood for 10 min using a dose as low as 1.05 mg/kg. Further, a polyethylene glycol (PEG) group at the 3′-end of DTI-33 did not reduce DTI potential. This is useful because PEG could possibly enhance the half-life of the molecule, however, the *in vivo* efficacy of the molecule was not tested.

Archemix has developed a formulation for another aptamer NU172 (also known as ARC2172, 5′CGCCTAGGTTGGG TAGGGTGGTGGCG3′), with slightly better pharmacokinetics and stability [25,125]. In Phase Ia trials, their novel formulation has demonstrated that 2 mg/kg bolus dose of NU172 could produce aPTT of 400 s and it returned to normal within ~ 10 min. In Phase Ib trial, the safety, tolerability and pharmacokinetics were studied in 24 healthy male volunteers. NU172 produced a dose-dependent increase in aPTT upto the highest infusion rate (6 mg/kg/hr) studied, which quickly returned to normal upon cessation of infusion. Importantly, no adverse events were observed [125]. NU172 could possibly find use in kidney

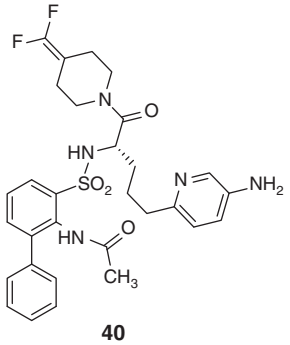
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37 (CDSO3) R = -OH, R′ = -H
38 (FDSO3) R = -OCH_3, R′ = -H
39 (SDSO3) R = -OCH_3, R′ = -OCH_3

Figure 16. Sulfated low molecular weight lignins as allosteric inhibitors of thrombin.



40
(SSR182289A)

Figure 17. SSR182289A which is a orally available non-peptide thrombin inhibitor developed as a nanospherical magnetic formulation which can be useful for localized therapy.

dialysis and surgical procedures, such as percutaneous coronary intervention, stent placement surgery, cardiopulmonary and coronary artery bypass surgery, angioplasty and other cardiovascular surgery. NU172 is currently being prepared for a Phase II clinical trial for use in patients undergoing off-pump coronary artery bypass graft surgery (*Clinical Trials* Identifier: NCT00808964) [126].

2.6.2 *Sulfated low-molecular-weight lignins*

Lignins are naturally occurring poly-hydroxyl containing aromatic polymers found commonly in wood. Lignin monomers are predominantly connected to each other through either β-5 or β-*O*-4 linkages [127]. Because the hydroxy groups on lignins can be sulfated, it was hypothesized that sulfated lignins may serve as useful mimetics of polymeric heparin. At the same time, sulfated lignins may offer a more hydrophobic aromatic backbone structure for enhanced selectivity of interaction. A recent patent application by Virginia Commonwealth University has described the synthesis of sulfated cinnamic acid-based oligomeric lignins by the names of CDSO3 (**37**), FDSO3 (**38**) and SDSO3 (**39**)(**Figure 16**), based on whether the starting lignin monomer is either caffeic acid, ferulic acid or sinapic acid, respectively [128]. The sulfated lignin polymers were potent inhibitors of thrombin (IC_{50} < 100 nM) by a direct and noncompetitive manner and bind to exosite 2 of the protease [129-131]. The polymers also directly inhibit factor Xa (IC_{50} < 250 nM). The most potent of these molecules was CDSO3, which showed good potency in clotting assays (42 µg/ml for doubling PT and 13 µg/ml for doubling APTT). *ex vivo* thromboelastography and haemostatic analysis studies in whole blood also found CDSO3 to be only 20-fold less active as compared to enoxaparin [132]. The sulfated lignins were chemo-enzymatically prepared, which is typically challenging and result in heterogeneity (**Figure 16**).

3. Advances in formulations

Argatroban has poor solubility in water [133]. The commercial injection solution is prepared using 100 mg argatroban with 750 mg *D*-sorbitol and 1 g dehydrated alcohol to make up a 1 liter stock, which is then diluted to 1 mg/ml [134]. Tianjin Weijie Technology has tested different alcohols to increase its solubility and stability [135]. At least one dehydrated alcohol from among propylene glycol, glycerol and *D*-sorbitol increased the solubility and stability of argatroban.

A major drawback of argatroban injections is high levels of ethanol and low concentration. This means that to administer a desired amount of argatroban, large volume of water and ethanol is to be injected, which makes the procedure too long, especially for a patient with small body weight. Baxter has utilized acid to improve the solubility by forming *in situ* salt solution of argatroban [136]. Acetic acid, formic acid, phosphate acid and tartaric acid are able to increase the solubility from 0.8 to 10 mg/ml. The optimized solution with pH 4.5 – 5.5 is stable to heat and light for at least 24 months. Eagle Pharmaceutics has also tested different acid-argatroban combinations under different temperatures [137]. The preferred solution contains 1 mg/ml argatroban, lactobionic acid and methionine.

Besides this optimization approach, improvement in bioavailability and convenience of administration has been attempted using vehicles that protect and control the release of the drug. Nanoparticles have been considered as a powerful delivery system for specific targeting and stability against degradation. Adding magnetite (Fe_3O_4) to form magnetic nanoparticles enables accumulation at target sites under an external magnetic field [138,139]. Jiangsu Yeming Bio-Medical Technology has employed magnetic nanosphere as a new delivery system for thrombin inhibitor SSR182289A (**40**) (**Figure 17**) [140]. SSR182289A is a potent and highly selective non-peptide oral DTI [141]. Incorporation of SSR182289A into poly(d,l-lactide-*co*-glycolide) and magnetic particle Fe_3O_4 forms nanosphere with diameter < 8 µm using an external magnetic field. Nearly 87% SSR182289A in the

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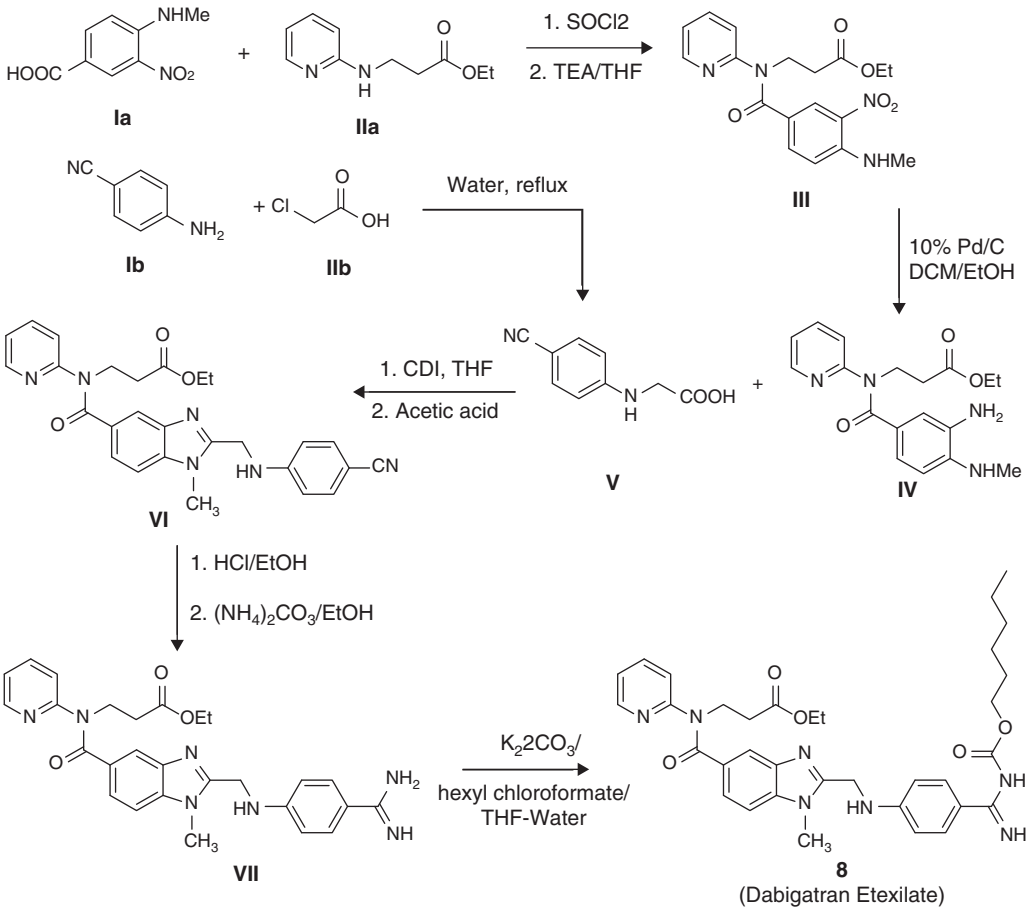


Figure 18. Original synthesis of dabigatran etexilate from intermediate VII.

nanosphere is released within 15 days. This could present excellent management of anticoagulant therapy for high risk patients. Although this technology has just been developed, it presents considerable promise for safe and efficient anticoagulant therapy that could be localized to a growing thrombus.

An improvement in the formulation of dabigatran etexilate has also been patented by Boehringer Ingelheim [142]. It is known that dabigatran etexilate methanesulfonic acid salt exists in different polymorphic forms, which could differ in stability, efficacy and solubility [143]. It is advisable to produce pharmaceutical composition consisting of essentially one polymorph. The invention describes a method of preparing a composition consisting of a single polymorphic form of dabigatran etexilate. First, core pellets made up of tartaric acid of size 0.4 to 0.6 mm are produced using binder acacia. These are then coated with a suspension of hydroxypropyl methylcellulose and talc to form isolated tartaric acid cores. Finally, an active ethanol suspension containing dabigatran etexilate methanesulfonate in the form of polymorph I is sprayed onto the isolated tartaric acid cores under controlled conditions (≤ 30°C). The coated pellets can then be filled into hydroxypropyl methylcellulose capsules. However, no data on benefits of using the single polymorph was provided in the disclosure.

4. Improvements in processes

The manufacture expense of drugs is a major determinant of clinic success, especially for anticoagulants because UFH and warfarin are inexpensive agents. Here, the recent improvements in the process of manufacture of already marketed or close-to-market DTIs are reviewed.

4.1 Argatroban purification

Argatroban (4) is derived from *L*-arginine. Published purification methods include water-alcohol reflux, HPLC, or specific argatroban antibody [144]. Yet, none of these can be applied at industrial scale due to low yields. Tianjin Weijie Technology has patented a large scale separation of argatroban by recrystallization [145], which still produces significant amount of side products containing the 21-(S)-isomer. Hainan Kangling Pharma has introduced a purification method using strong anion exchange resin absorption followed by recrystallization to achieve 99.4% purity [146].

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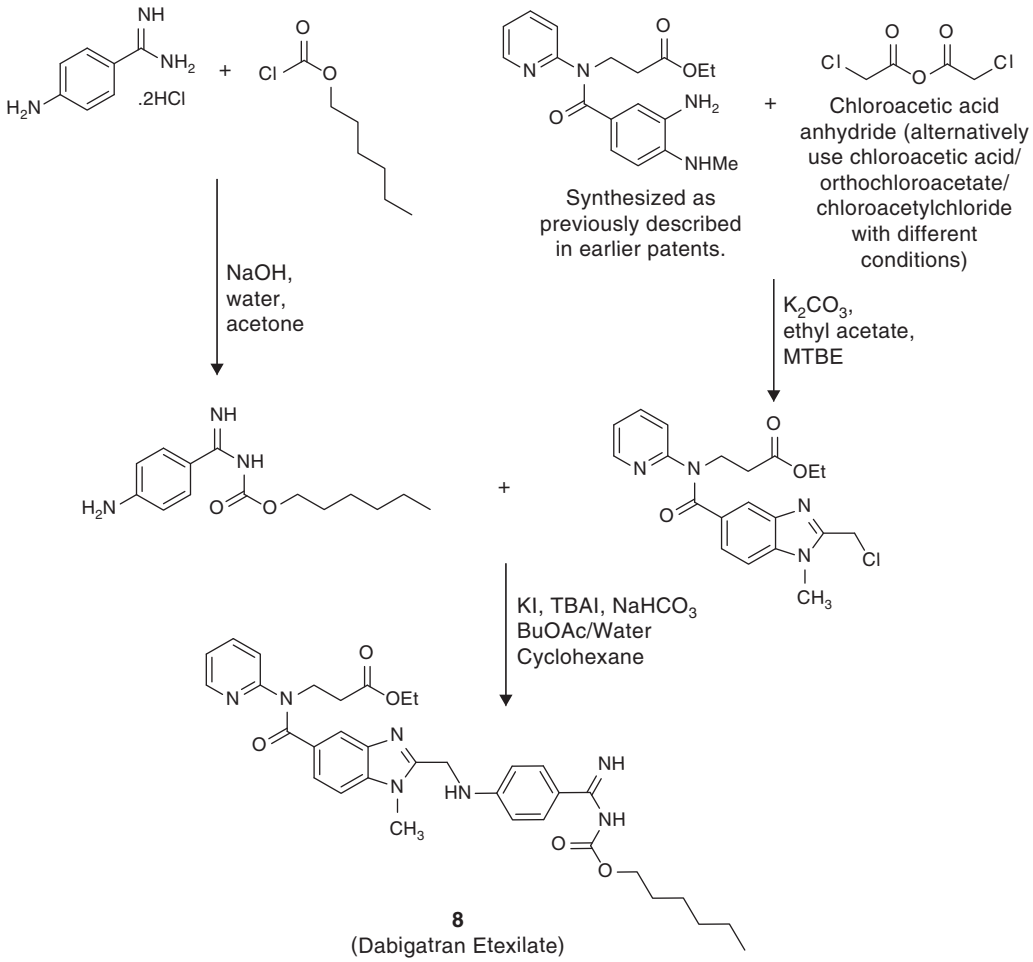


Figure 19. Newer method for the synthesis of dabigatran etexilate by Boehringer Ingelheim.

4.2 Production of dabigatran and dabigatran etexilate

The original synthesis of dabigatran etexilate, proposed by Boehringer Ingelheim in 1998 [147], is scalable, but suffers from low efficiency arising from expensive and complicated purification steps. In particular, the amidine-containing intermediate **VII** (**Figure 18**) was difficult to synthesize as well as isolate. The original conversion of **VI** – **VII** occurs in two steps without any purification of the imidate intermediate. The first step involves acidification using HCl in ethanol, while the subsequent step requires large amounts of ammonium carbonate (~ 11 equiv.) to neutralize the excess acid. The large amount of salt so produced is difficult to remove and contaminates the product, which affects subsequent steps and yield. Several advancements in these steps have been patented recently.

4.2.1 *Modifications to original dabigatran synthesis*

A patent application suggests the purification of the imidate intermediate as a hydrochloride salt prior to reaction with ammonia [148]. The method also outlines the ability to reduce the equivalents of ammonia used to 3 – 5 equivalents (instead of ~ 11), which reduces the salt formation nearly 5-fold in the formation of amidine **VII**. In 2010, Lustig *et al*. produced the intermediate **VII** as a monohydrochloride salt, which could be dissolved in ethanol for easy recrystalization in high purity (90 – 95%) without the need for chromatographic purification [149]. Further treatment of this monohydrochloride with HCl gave the dihydrochloride salt of **VII** in very high purity (96 – 98%). Another recent patent by Zentiva claimed further improvements to the process by forming the hydrochloride salt of **III**, which can be purified by recrystalization from ethanol–acetonitrile mixture (> 99%) [150]. This also facilitated its conversion to **IV** in good purity (> 95%) using sodium dithionite without the use of palladium over carbon, thus eliminating the need for high temperatures and pressure. The patent also discloses that the oxalic acid salt of **VI** allows recrystalization of the intermediate in reasonable purity (80 – 90%) using ethyl alcohol. In addition, the use of bromoacetic acid, in place of chloroacetic acid (**IIb**), to form **V** increased the yield to 85 – 90% as compared to 49% due to bromide ion being a better leaving group.

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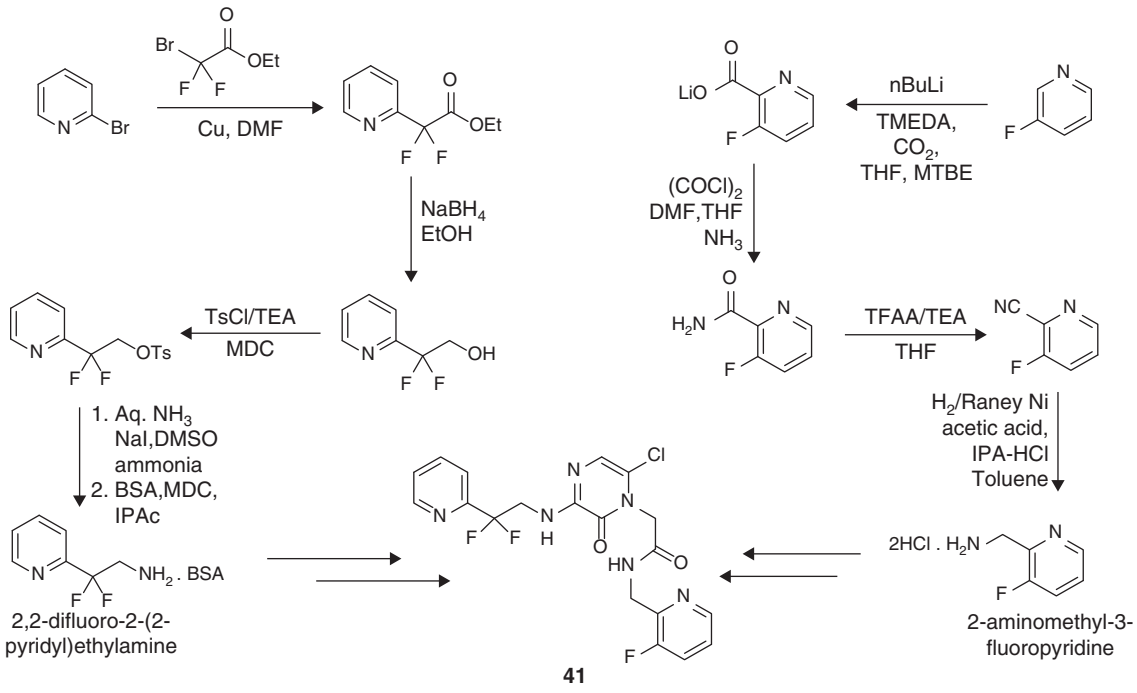


Figure 20. New process for synthesis of orally bioavailable thrombin inhibitor 41.

4.2.2 *New synthesis of dabigatran etexilate*

Boehringer Ingelheim has disclosed a completely different strategy for industrial scale synthesis of dabigatran in a recent patent application (**Figure 19**) [151]. The synthesis uses readily available starting agents, reduces the number of steps and use of acids (which are expensive to dispose), favors phase transfer catalysis in the formation of the final product (which avoids the use of coupling reagents) and obviates the need for chromatographic separations. This results in a significantly more efficient synthetic strategy. This strategy seems to be promising; however, the patent application discloses reactions at 30 g scale only, which begs the question on scalability at industrial levels.

4.3 Preparation of pyrazinone thrombin inhibitors

Over a decade ago, Merck patented a lead DTI 3-fluoro-2-pyridylmethyl-3-(2,2-difluoro-2-(2-pyridyl)ethylamino)-6-chloro-pyrazin-2-one-1-acetamide (**41**) [152], which displayed a K_i of 5.2 nM and selectivity of 38,800-fold over trypsin. The inhibitor also demonstrated good oral bioavailability in animals (66% in dogs and African green monkeys) and a good plasma half-life of 6.6 h in dogs at a dose of 1 mg/kg [153]. Further, **41** was found to be stable to human liver microsomes and hepatocytes, which led to the prediction of its low clearance in man. Thus **41** was very attractive drug-like DTI. However, the synthesis of this agent involved several chromatographic separations making it challenging at large scales. Diakron Pharmaceuticals has recently obtained a process patent for producing **41** in bulk quantities **Figure 20** [154]. The process involves ready to use and cheap raw materials allowing the industrial scale-up of **41** through two key intermediates, 2-aminomethyl-3-fluoropyridine and 2,2-difluoro-2-(2-pyridyl)ethylamine. The patent highlights that scaled up production are made possible by eliminating chromatographic purifications thus reducing cost and enhancing efficiency (**Figure 20**).

5. Expert opinion

In addition to the therapeutic efficiency as an inhibitor of coagulation, an ideal anticoagulant would be one that possesses a broad therapeutic window, is devoid of bleeding risk, does not require for frequent coagulation monitoring, is orally bioavailable, can be effectively neutralized through an antidote treatment, lacks hepatotoxicity, does not pose any other adverse effects and is relatively inexpensive. No anticoagulant developed so far possess all of these properties and hence the quest for an ideal molecule continues unabated.

Several advances have been made in the past decade or so. The long established view that blocking thrombin activity would yield a potent anticoagulant continues to remain attractive [155]. In particular, the FDA approval of orally administered DTI dabigatran, especially for prophylactic use, has re-energized the search for better DTIs. Yet, the effort to uncover a truly ideal anticoagulant is likely to be challenging.

Over 15 years ago, the previous report on thrombin inhibitors in this journal was published and the majority of

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inhibitors discussed were targeting the active site [132]. Today, our understanding of the roles thrombin plays in the clotting system has vastly expanded [22]. Thrombin's catalytic activity is not just a simple function of its active site but also is dependent on the state of its allosteric sites [22-28]. Indeed, this patent review suggests that the trend in the development of DTIs is changing. Complex macromolecules, allosteric modulators and also multiply functionalized active site-targeted inhibitors are being pursued. A general consensus to be drawn from this search is that thrombin's multi-functional role demands a complex inhibitor design with sensitive regulatory features. In fact, a particularly novel paradigm that appears to have not been pursued is the local cue dependent shift in a molecule's function from a procoagulant state to an anticoagulant state, in a manner similar to thrombin.

Besides DTIs, synthetic ultra-low-molecular-weight heparins are also being pursued to combat thrombosis without having the side effect associated with full-length heparins [156]. A couple of factor Xa inhibitors, rivaroxaban and apixaban, have been recently approved for clinical use [157] and intrinsic pathway enzymes are starting to be targeted to achieve antithrombotic effects without bleeding complications [158,159]. Factor IXa inhibitor TTP889 is currently being investigated as an oral prophylactic antithrombotic agent and has cleared Phase II trials (*Clinical Trials Identifier:* NCT00119457) [43,160,161]. Likewise, factor XIa inhibitors are also being developed [162-164]. Yet, it is important to recognize that response variability and the individual needs may require several nearly ideal anticoagulants to be developed for patients around the world. Many candidates outlined in this review are attempting to achieve this, yet there is considerable room for further exploration.

Declaration of interest

This work was supported by grants HL090586 and HL107152 from the National Institutes of Health to URD.

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